Understanding Keyword Searches

Exploring with Apollo

Brittany Dueker

Hello!

My name is Brittany, I use she/her pronouns, and for those of you who can't see me, I am a white woman with light brown hair wearing a brown sweater.

Today, I'll be talking about keyword searches, specifically in the library management software Apollo.

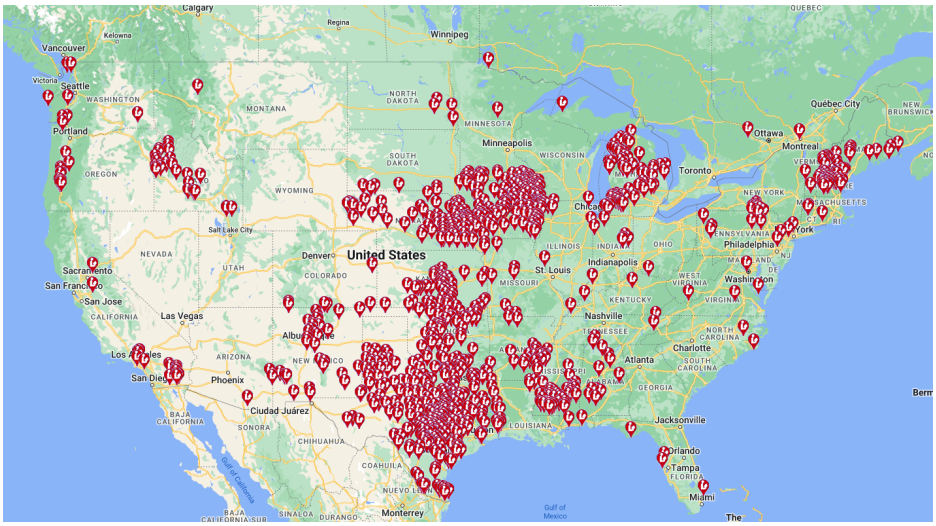
Content and Goals

- Describe what fields in Apollo are keyword searchable
- Identify problems with using keyword searches
- Give examples of how to use this information to improve your catalog searches

This presentation is designed for a conference for the Iowa Library Association's annual conference. The expected audience is library professionals and other conference attendees, many who may not have a formal library education.

My goals with this talk are to 1) describe keyword searchable fields in Apollo, 2) identify problems that occur when using keyword searches, and 3) give you a few examples of ways for you to improve your catalog searches

30 % of public libraries in Iowa use Apollo

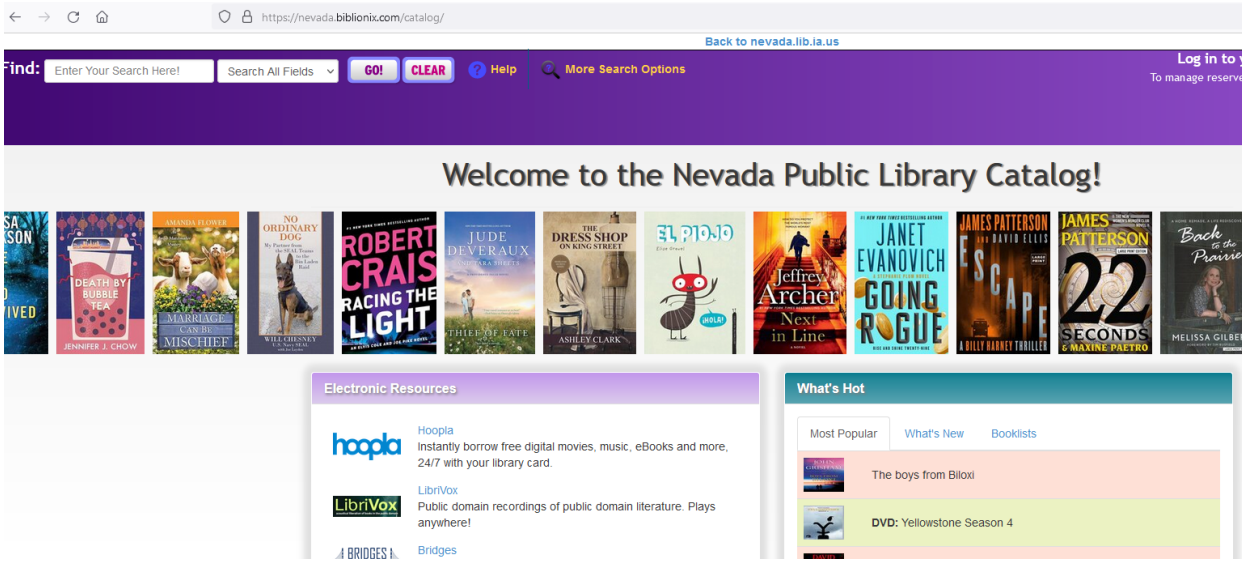


Map courtesy of librarytechnology.org ; data from libraries.org

Apollo is a library management software created by Biblionix, designed for public library collections of fewer than 300,000 items. It is preferred by many smaller libraries as it is less expensive and has an easier learning curve than other software with more robust features.

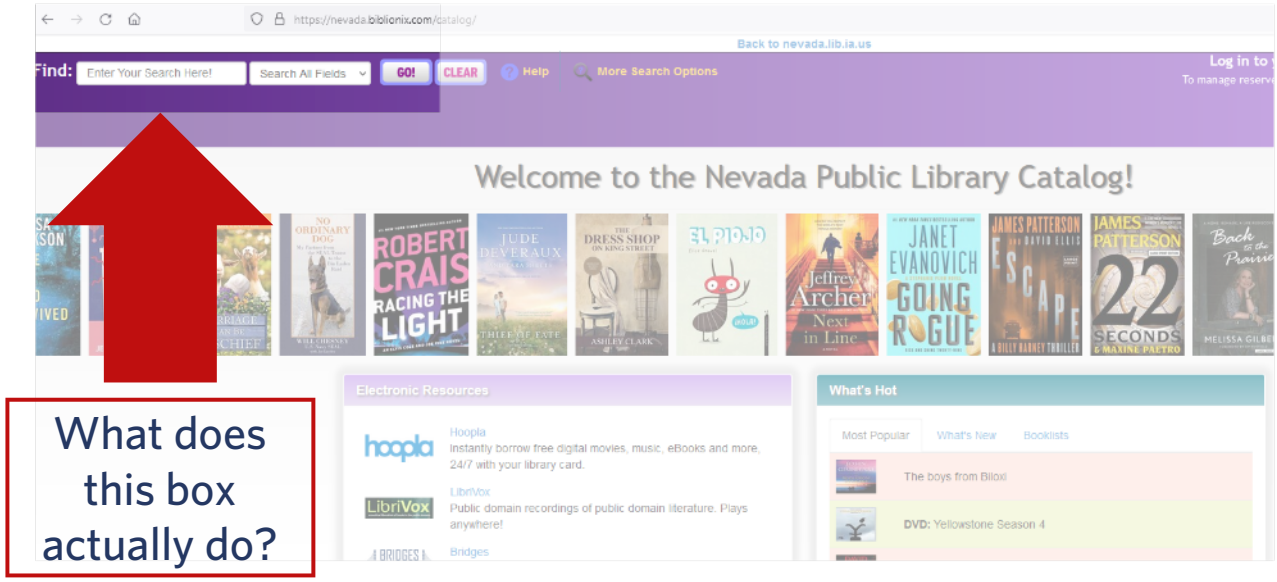
The map on this slide shows where the public libraries that use Apollo are located. It's very popular in Texas, Nebraska, Michigan, New Hampshire, and Iowa. In fact, 30% of public libraries in Iowa use Apollo.

What is searchable?



One of those libraries is the public library in Nevada, Iowa. This is the Nevada Public Library's main catalog page, or at least a portion of it. When anyone goes to this page, myself included, generally the first thing they do is type in a keyword in the search box at the top.

What is searchable?



What does this box actually do?

But what does this box actually do?

What does the user manual say?

A keyword search returns bibliographic records that contain all of the entered terms in any of the following fields:

- Author
- Call Number
- Medium
- Notes
- Subjects
- Title
- ISBN

Typing in multiple words gives you an AND search of all the words.

According to the user manual, when a keyword search is done, the terms are searched for in the following fields:
Author
Call Number
Medium
Notes
Subjects
Title
ISBN.

It then tells you that using more than one keyword will cause the system to search for the words as if they had AND in between them.

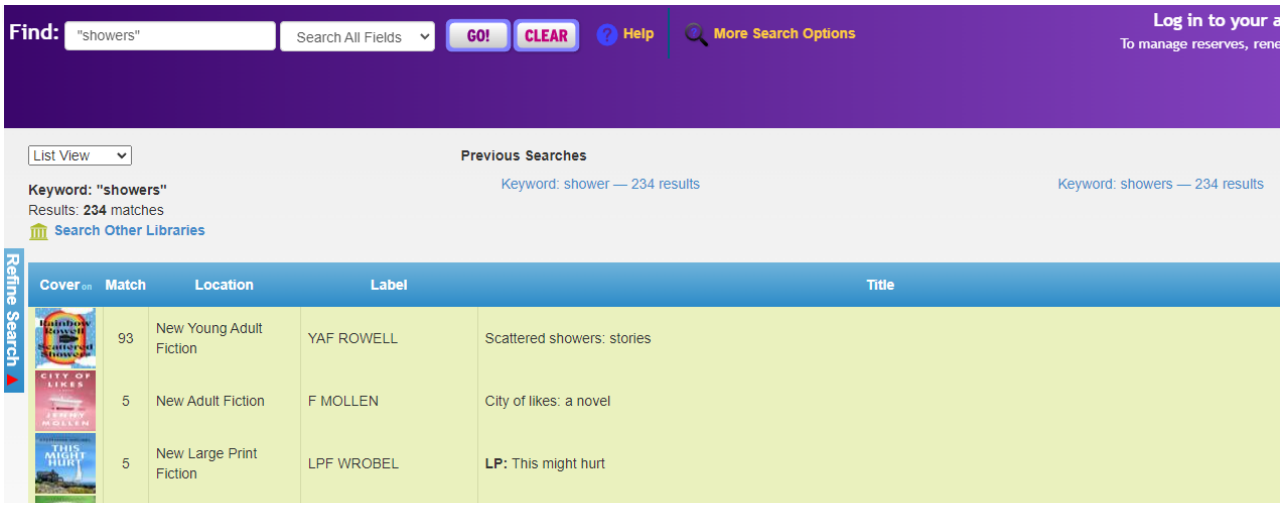
What does the user manual say?

For every keyword entered, several alternate versions of that keyword will also be searched.

- Singular or plural
- Hyphens removed, replaced with spaces, or words split at likely hyphenation points
- Possessive, non-possessive, or with apostrophes removed
- Digits converted to words and vice versa
- Cardinal numbers to ordinal numbers and vice versa

The user manual also says that the software searches for alternate versions of keywords, including singular or plural versions, various hyphenations and possesives. It converts digits to words and cardinal numbers to ordinal numbers and vice versa. So a search for the number 3 will also search for the word three, the word third, and the numerical abbreviation 3rd.

Searching for "showers"



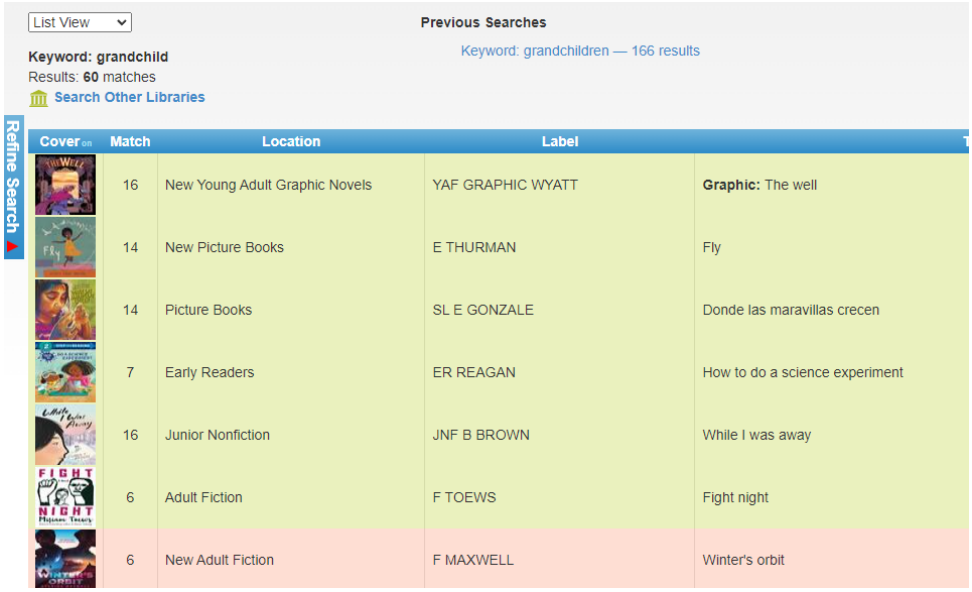
I experimented a little bit with this, since the user manual doesn't give examples or go into any more detail. What I found was kind of interesting.

Using quotation marks doesn't prevent the keywords from being searched with alternate versions.

"Shower" (with quotes) and shower (without quotes) return the same results, as shown with this picture of a catalog search. Note the search history that shows that there are 234 results either way.

"Baby shower" (with quotes) and baby shower (without quotes) do return different results, so quotation marks help to search for phrases, but "baby shower" and "babies showers" return the same results.

Searching for "grandchildren"



List View

Previous Searches

Keyword: grandchildren — 166 results

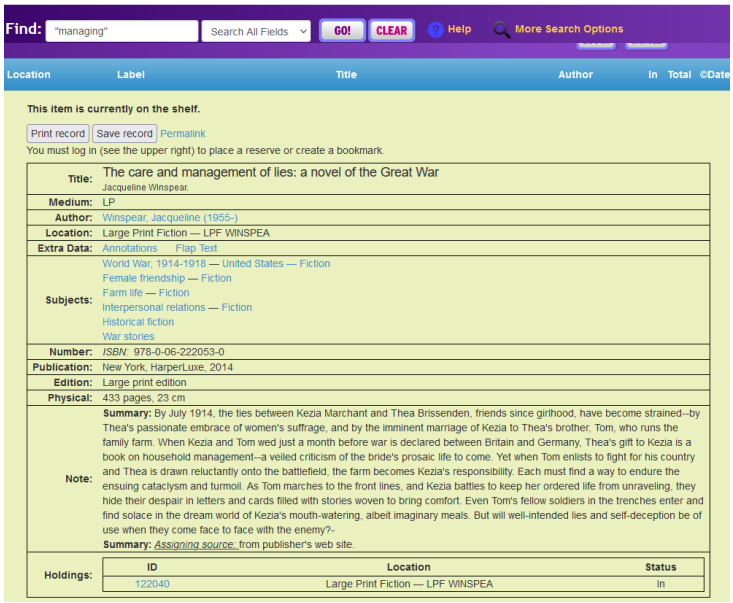
Keyword: grandchild
Results: **60** matches
Search Other Libraries

Refine Search

Cover	Match	Location	Label	
	16	New Young Adult Graphic Novels	YAF GRAPHIC WYATT	**Graphic:** The well
	14	New Picture Books	E THURMAN	Fly
	14	Picture Books	SL E GONZALE	Donde las maravillas crecen
	7	Early Readers	ER REAGAN	How to do a science experiment
	16	Junior Nonfiction	JNF B BROWN	While I was away
	6	Adult Fiction	F TOEWS	Fight night
	6	New Adult Fiction	F MAXWELL	Winter's orbit

Despite the manual saying that the singular and plural versions of a word get searched, it doesn't seem to work quite correctly. Whatever list that Apollo is using is incomplete, since searching for grandchildren produces a different number of results and also different titles than grandchild, as shown in this catalog search for grandchild. Note that there are only 60 results for grandchild and 166 results for grandchildren, when the searches should be equivalent.

Searching for "managed"



Find: "managing" Search All Fields GO! CLEAR Help More Search Options

Location	Label	Title	Author	In	Total	©Date

This item is currently on the shelf.

Print record Save record Permalink

You must log in (see the upper right) to place a reserve or create a bookmark.

Title:	The care and management of lies: a novel of the Great War Jacqueline Winspear.
Medium:	LP
Author:	Winspear, Jacqueline (1955-)
Location:	Large Print Fiction — LPF WINSPEA
Extra Data:	Annotations Flap Text
Subjects:	World War, 1914-1918 — United States — Fiction Female friendship — Fiction Farm life — Fiction Interpersonal relations — Fiction Historical fiction War stories
Number:	*ISBN:* 978-0-06-222053-0
Publication:	New York, HarperLuxe, 2014
Edition:	Large print edition
Physical:	433 pages, 23 cm
Note:	**Summary:** By July 1914, the ties between Kezia Marchant and Thea Brissenden, friends since girlhood, have become strained--by Thea's passionate embrace of women's suffrage, and by the imminent marriage of Kezia to Thea's brother, Tom, who runs the family farm. When Kezia and Tom wed just a month before war is declared between Britain and Germany, Thea's gift to Kezia is a book on household management--a veiled criticism of the bride's prosaic life to come. Yet when Tom enlists to fight for his country and Thea is drawn reluctantly onto the battlefield, the farm becomes Kezia's responsibility. Each must find a way to endure the ensuing cataclysm and turmoil. As Tom marches to the front lines, and Kezia battles to keep her ordered life from unraveling, they hide their despair in letters and cards filled with stories woven to bring comfort. Even Tom's fellow soldiers in the trenches enter and find solace in the dream world of Kezia's mouth-watering, albeit imaginary meals. But will well-intended lies and self-deception be of use when they come face to face with the enemy?- **Summary:** *Assigning source:* from publisher's web site.

Holdings:

ID	Location	Status
122040	Large Print Fiction — LPF WINSPEA	In

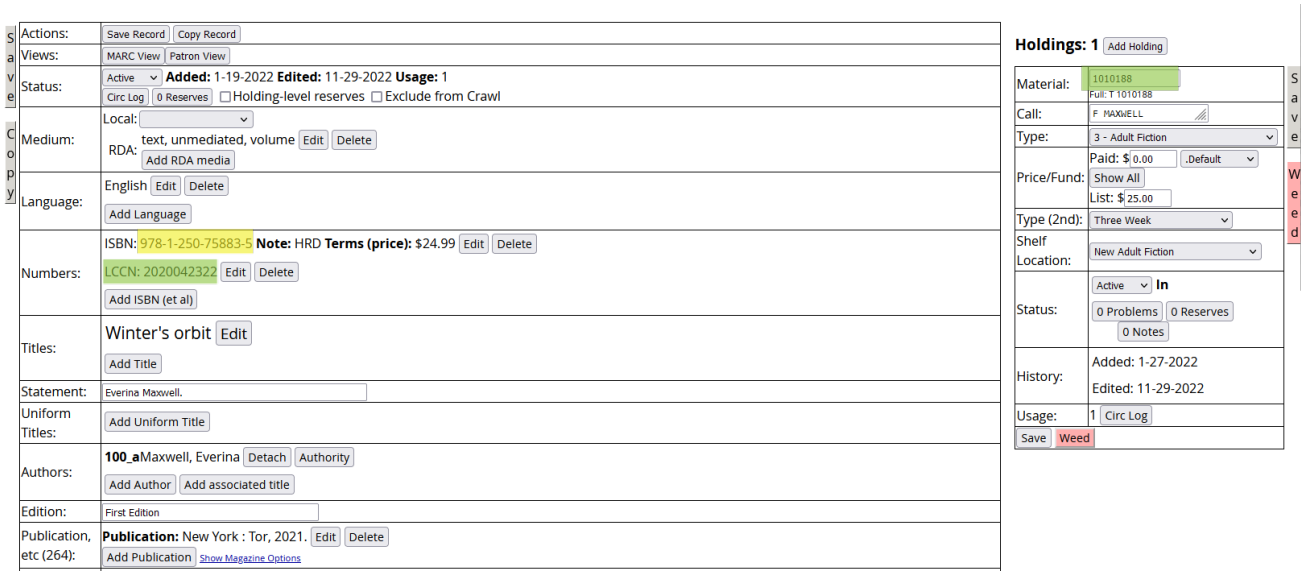
In addition to missing certain words, Apollo doesn't limit its variant terms to just plurals.

On this slide is an image of a catalog record found by searching for the word "managing," despite the word managing not being in this record. The word "management" does appear in both the title and the summary, though.

It works in the other direction, too. A search for "manage" brings up titles with records that contain the words "managed," manages, "management," and "managing."

It's not clear how Apollo chooses its root words or endings, but the variability exists to some extent.

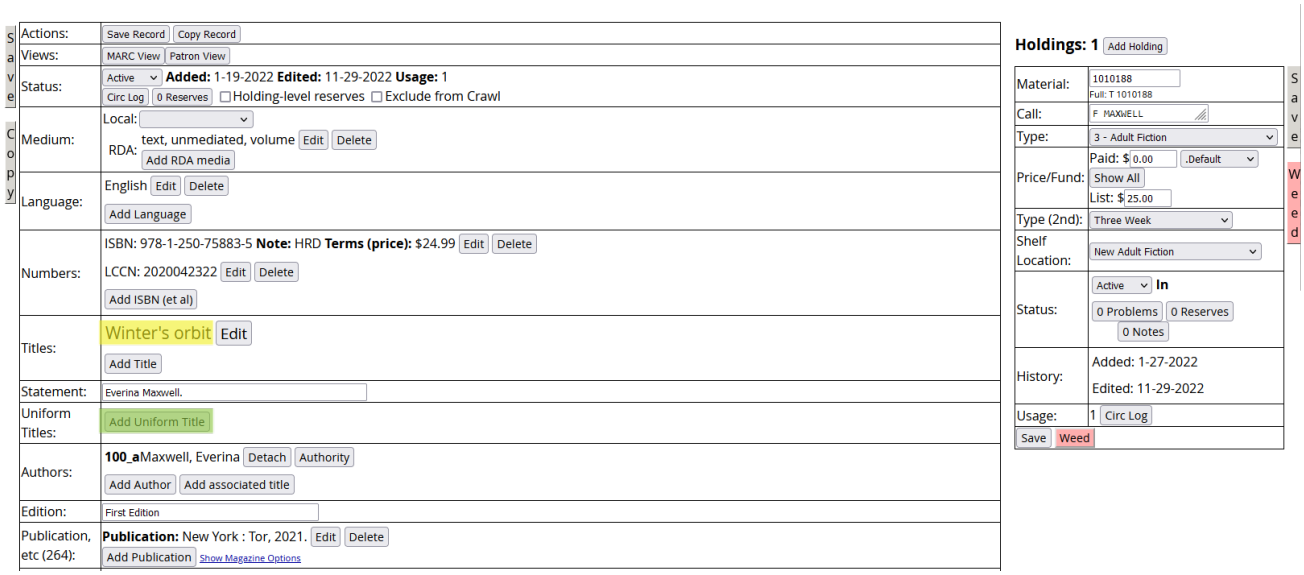
Behind the Scenes



What I've shown so far has all been public-facing. The catalog looks very different on the back end. This is a catalog record as seen from the cataloger's view. I've highlighted some of the keyword-searchable fields. Yellow fields are mentioned in the manual as keyword-searchable and green fields are not mentioned but are also searchable. ISBN is yellow; LCCN and Material # are green.
In fact, any number in the Numbers field is keyword searchable, including video recording numbers and UPCs.

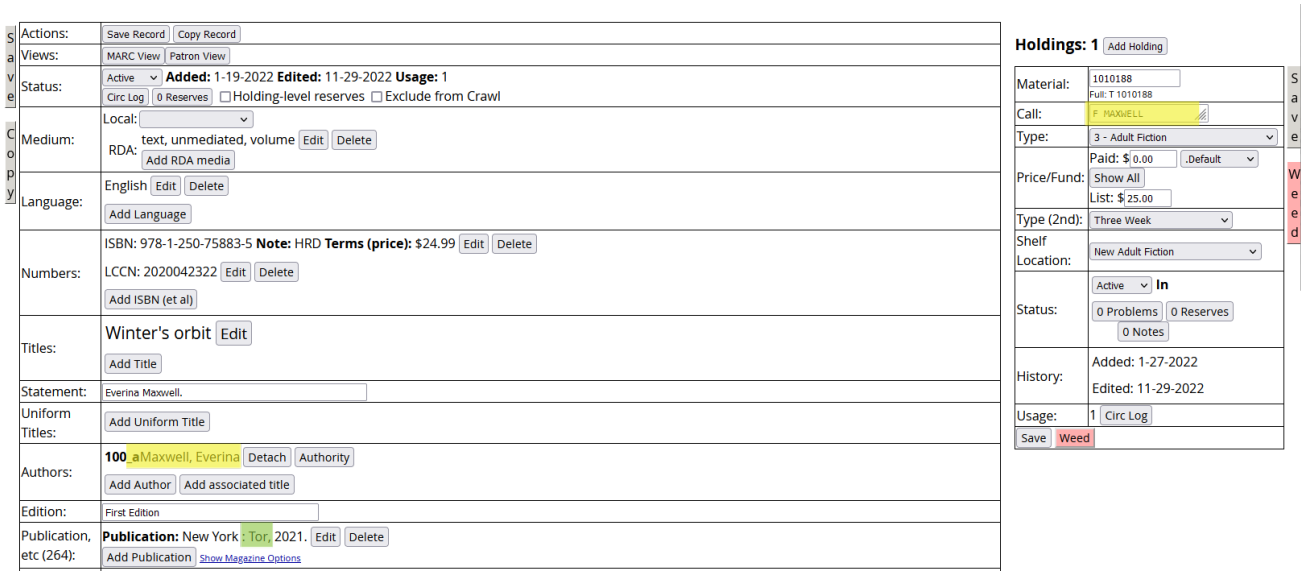
Material number is less helpful for a general search, but it's an easy way to access a catalog record of a book you have in-hand so that you can use the catalog find similar titles.

Behind the Scenes



Predictably, title is also a keyword-searchable field. You can also search for uniform titles, if your library has items in multiple languages. Uniform titles are treated as subject headings in the public-facing side of the catalog and can be found in the subject heading section there.

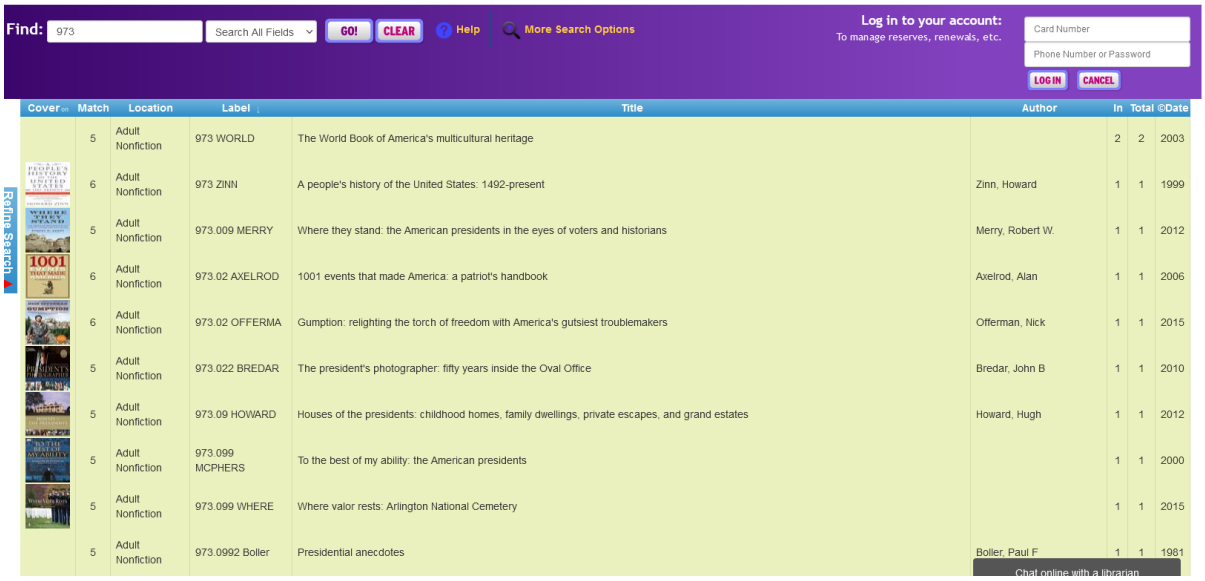
Behind the Scenes



The author field is keyword-searchable, but the statement field is not.
Oddly not mentioned in the manual, the publisher field is keyword-searchable. Most publishers are genre-specific, so this might be an excellent way to do a general search, if you're familiar with the publishers' specialties.

Call numbers are also searchable, although it's a little tricky. Using the location part of the call number ("F" here) by itself works, as does any other word that is part of the call number, as long as it matches exactly.

Searching for "973"



Cover	Match	Location	Label	Title	Author	In	Total	©Date
	5	Adult Nonfiction	973 WORLD	The World Book of America's multicultural heritage		2	2	2003
	6	Adult Nonfiction	973 ZINN	A people's history of the United States: 1492-present	Zinn, Howard	1	1	1999
	5	Adult Nonfiction	973.009 MERRY	Where they stand: the American presidents in the eyes of voters and historians	Merry, Robert W.	1	1	2012
	6	Adult Nonfiction	973.02 AXELROD	1001 events that made America: a patriot's handbook	Axelrod, Alan	1	1	2006
	6	Adult Nonfiction	973.02 OFFERMA	Gumption: relighting the torch of freedom with America's gutsiest troublemakers	Offerman, Nick	1	1	2015
	5	Adult Nonfiction	973.022 BREDAR	The president's photographer: fifty years inside the Oval Office	Bredar, John B	1	1	2010
	5	Adult Nonfiction	973.09 HOWARD	Houses of the presidents: childhood homes, family dwellings, private escapes, and grand estates	Howard, Hugh	1	1	2012
	5	Adult Nonfiction	973.099 MCPHERS	To the best of my ability: the American presidents		1	1	2000
	5	Adult Nonfiction	973.099 WHERE	Where valor rests: Arlington National Cemetery		1	1	2015
	5	Adult Nonfiction	973.0992 Boller	Presidential anecdotes	Boller, Paul F	1	1	1981

Chat online with a librarian

Part of nonfiction call numbers, at least in my library, is a Dewey number. A keyword search for a Dewey number is not intuitive.
Here is an image of part of a catalog search for 973, which gives back items with any call number starting with 973, including numbers with decimal places like 973.09 and 973.0992.

Searching for "973.09"

Find: 973.09 | Search All Fields | GO! | CLEAR | Help | More Search Options

Log in to your account:
To manage reserves, renewals, etc.
Card Number
Phone Number or Password
LOG IN | CANCEL

List View

Previous Searches
Keyword: 973 — 385 results

Keyword: 973.09
Results: **11** matches
Search Other Libraries

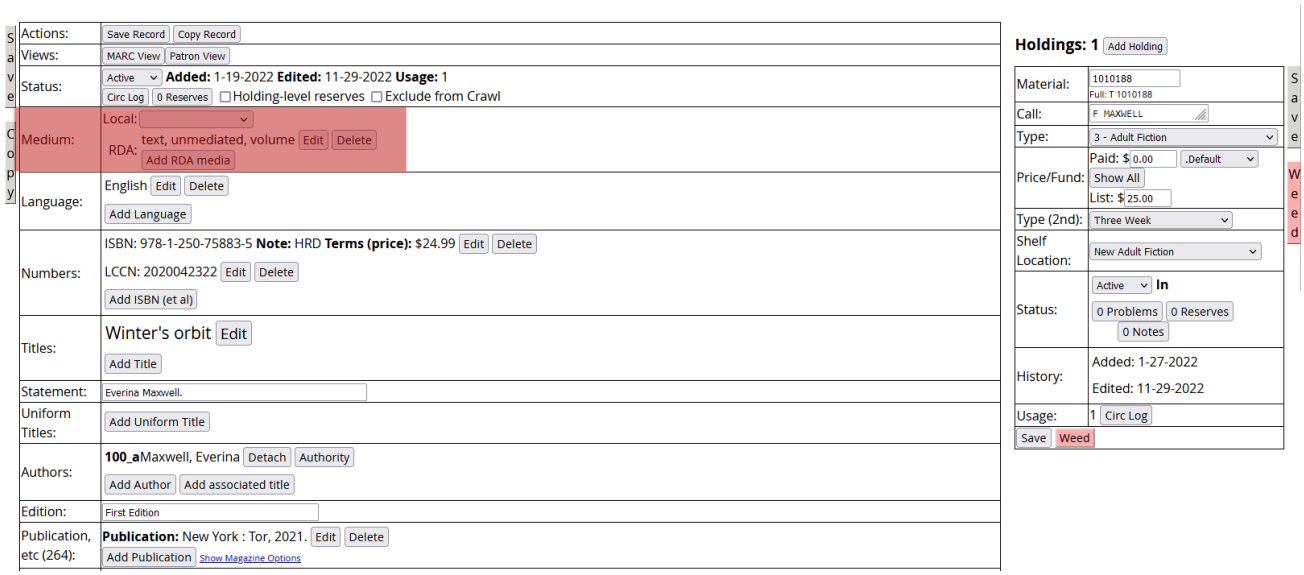
Refine Search

Cover	Match	Location	Label	Title	Author	In	Total	©Date
	89	Adult Nonfiction	973.09 HOWARD	Houses of the presidents: childhood homes, family dwellings, private escapes, and grand estates	Howard, Hugh	1	1	2012
	80	Adult Nonfiction	973.09 CAROLI	First ladies: from Martha Washington to Michelle Obama	Caroli, Betty Boyd	1	1	2010
	80	Adult Nonfiction	973.09 RICKS	First principles: what America's founders learned from the Greeks and Romans and how that shaped our country	Ricks, Thomas E	1	1	2020
	78	Adult Nonfiction	973.09 BROWER	Team of five: the presidents club in the age of Trump	Brower, Kate Andersen	1	1	2020
	64	Adult Nonfiction	973.09 GRADDY	The Smithsonian First Ladies Collection	Graddy, Lisa Kathleen	1	1	2014
	64	Adult Nonfiction	973.09 GOODWIN	Leadership in turbulent times	Goodwin, Doris Kearns	1	1	2018

Search Other Libraries

When you do a keyword search for 973.09, it gives back only items with call numbers that are exactly 973.09 and doesn't include items with longer call numbers like 973.0992.

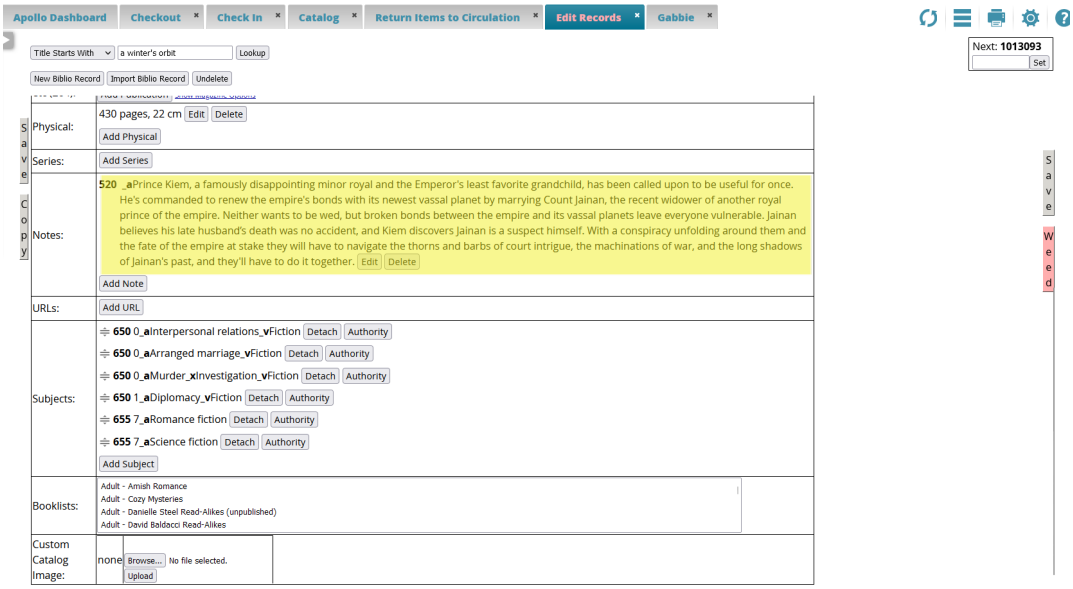
Behind the Scenes



Back to the back-end catalog view, I've highlighted the Medium field, which describes the kind of material (such as large print or audiobook) and the RDA information. Despite being mentioned in the manual, the medium field is not keyword-searchable.

There is a filter in the public catalog search menu, for the local medium, though.

Behind the Scenes



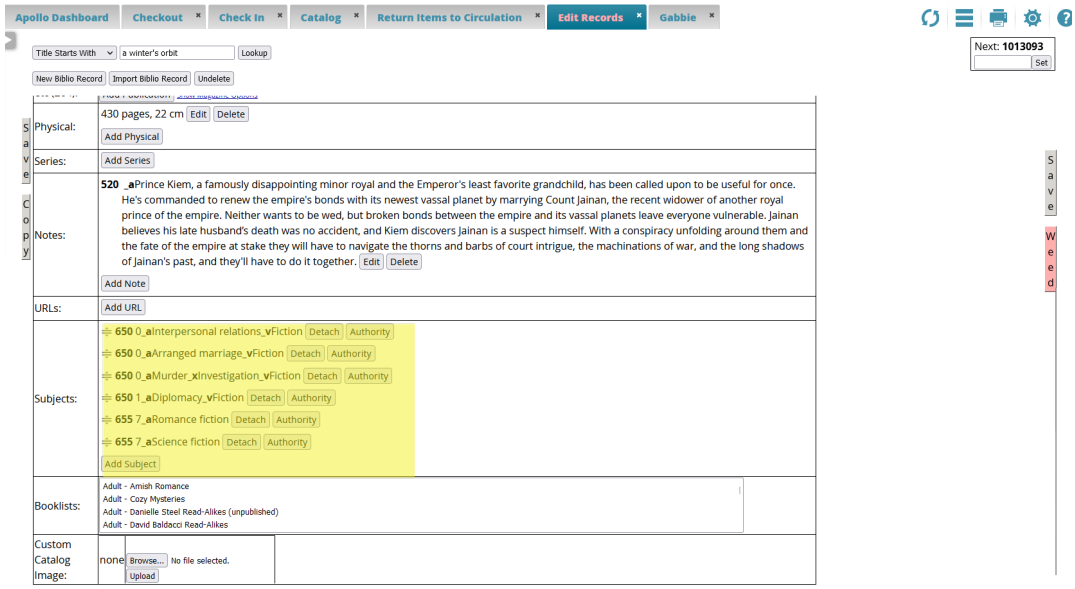
If we scroll down to the bottom of the back end of the catalog record, we find that, of course, summaries are keyword-searchable. Summaries aren't always consistent in the information that they contain and can often be the reason that a search provides confusing results. Potentially having additional information here is useful, but not when you're looking for something specific.

A search for Tolkien

Title:	The black elfstone: the fall of Shannara Terry Brooks.
Series Title:	The fall of Shannara ; (#1)
Author:	Brooks, Terry
Location:	Adult Fiction — F BROOKS FALL #01
Extra Data:	Annotations Biography Excerpt
Subjects:	Fantasy fiction FICTION / Action & Adventure FICTION / Fantasy / Epic FICTION / Science Fiction / Adventure Shannara (Imaginary place) — Fiction
Number:	*ISBN:* 978-0-553-39148-0 *LCCN:* 2017015642
Publication:	New York, Del Rey, [2017]
Edition:	1st Ed.
Physical:	318 p., 25 cm
Note:	**Summary:** "The first book of the triumphant and epic four-part conclusion to the Shannara series, from one of the all-time masters of fantasy. Across the Four Lands, peace has reigned for generations. But now, in the far north, an unknown enemy is massing. More troubling than the carnage is the strange and wondrous power wielded by the attackers--a breed of magic unfamiliar even to the Druid order. Fearing the worst, the High Druid dispatches a diplomatic party under the protection of the order's sworn guardian, Dar Leah, to confront the mysterious, encroaching force and discover its purpose. But another crucial journey is being undertaken. Exiled onetime High Druid Drisker Arc has been living in quiet seclusion, far from the politics and power struggles of his former life, until two brutal attacks by would-be assassins force him to seek out an infamous murder-for-hire guild--and find the hidden enemy who has marked him for death. At his side is Tarsha Kaynin, a young woman gifted with the wishsong and eager to be schooled in its formidable power by a master. She, too, is pursuing a mission: to locate her wayward brother, whose own magic has driven him to deadly madness and kindled his rage for vengeance. against his sister. In their darkest hours, facing dangerous adversaries, the lives and quests of Dar Leah, Drisker Arc, and Tarsha Kaynin will be inextricably drawn together. And the challenges each confronts will have resounding consequences for the future of the Four Lands. Praise for Terry Brooks "The Sword of Shannara is an unforgettable and wildly entertaining epic, animated by Terry Brooks's cosmically generative imagination and storytelling joy."--Karen Russell, New York Times bestselling author of Swamplandia! "If Tolkien is the grandfather of modern fantasy, Terry Brooks is its favorite uncle."--Peter V. Brett, New York Times bestselling author of The Skull Throne "I can't even begin to count how many of Terry Brooks's books I've read (and reread) over the years. From Shannara to Landover, his work was a huge part of my childhood."--Patrick Rothfuss, New York Times bestselling author of The Name of the Wind "Terry Brooks is a master of the craft and a trailblazer who established fantasy as a viable genre. He is required reading."--Brent Weeks, New York Times bestselling author of The Night Angel Trilogy "The Shannara books were among the first to really capture my imagination. My daydreams and therefore my stories will always owe a debt to Terry Brooks."--Brandon Mull, #1 New York Times bestselling author of the Beyonders series and the Fablehaven series"-- **Summary:** *Assigning source:* Provided by publisher. **Summary:** "The first book of the triumphant and epic four-part conclusion to the Shannara series, from one of the all-time masters of fantasy. The Four Lands has been at peace for generations, but now a mysterious army of invaders is cutting a bloody swathe across a remote region of the land. No one knows who they are, where they come from, or what they are after--and most seem

As an example, searching for "Tolkien" brings up all of the the materials written by JRR Tolkien and his son and grandson. It also brings up the book "The Black Elfstone", which has nothing to do with Tolkien (except that it's a fantasy book). On this slide is an image of the public facing catalog record for the book. I've highlighted in blue where Tolkien is mentioned in the notes, which happens to be a quote from another author. I've highlighted in red five other authors, three other book titles, and three other book series that appear in this record. A search for any of these keywords would bring up this particular Terry Brooks novel. That's potentially useful, if you are looking for read-alikes. I mostly find it obnoxious and clutter-y, so as a cataloger, I remove these kinds of things when I see them.

Behind the Scenes



In the back-end of the record again, I've highlighted the last keyword-searchable field, the subjects field. While it depends on how items in your library are cataloged, using a keyword search with subject heading terms can give you a nice broad range of materials on a particular topic or in a particular genre. Using a subject-heading search will likely be a better way to find things, but if your catalog is like mine (i.e. in desperate need of a tidy of the authority records), a keyword search might provide a wide selection of relevant titles.

Practical Application

- Try a few searches in your library's software. Do you get the results you expect?
- What does the user guide say? Is it correct?
- If you catalog:
  - How do your coworkers and patrons use the catalog?
  - What can you do to improve searchability?
  - Are there quirks you've noticed that you could share with your coworkers?

Although I've spent this time talking about Apollo software specifically, there are a few ways you can apply this knowledge to any catalog. First, try a few searches in your catalog. Are the results what you expect? Are there oddities in your results that you can't explain? If so, what does the user guide say? Test it out. Does the search work according to the instructions?

If you do any library cataloging, consider how your coworkers and patrons use the catalog. Is there anything you can do to improve searchability? Are there any quirks you've noticed that you can share?

Questions? Comments? Contact Me!

CITY OF NEVADA LIBRARY

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Thank you for your time and attention! Feel free to contact me at brittany-dueker@uiowa.edu for copies of the slides or with any questions or comments. Have a wonderful day!